

Abdelrhman Saeed

AI Engineer

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EDUCATION

Arab Academy for Science, Technology, and Maritime

Bachelors in Artificial Intelligence, GPA: 3.2/4.00, Dean's List

Alexandria, Egypt

Expected Aug 2025

- Concentration: Robotics and Intelligent Systems
- Relevant Coursework: DS & Algorithms, OOP, ML, DL, RL, CV, NLP, HPC, Cloud Computing, Robotics, Database Systems Design, Web Programming, Ai In Web Programming, Mobile Computing

EXPERIENCE

Machine Learning Course - ITI

Trainee

Alexandria, Egypt

Aug 2024 – Sep 2024

- Completed a 60-hour machine learning course covering supervised and unsupervised learning through practical projects.

ICPC Competitive Programmer

Team Leader & Contestant

Alexandria, Egypt

Jun 2023 – Jun 2024

- Led college team in national ICPC contests.
- Finalist in ECPC and ACPC. Ranked 1st in college in both ECPC finals. Achieved first solve in 2024.

RoboCup@Home EDU Workshop

AI Robotics Specialist

Alexandria, Egypt

Feb 2023 – Mar 2023

- Programmed the Jupiter robot with C++ and ROS. Applied vision techniques to enable interaction.

Competitive Programming

Active Problem Solver

Online

Jun 2021 – Present

- Codeforces: Achieved Expert-level rating of **1616** (max), solved over **1700** rated problems—placing in the 1600–1899 “Expert” tier on a platform that challenges 1.7M+ users

PROJECTS

B.E.M.O (Graduation Project) — Python, LangChain, Tavily, OpenCV, FastAPI

Jul 2024 – Present

- Developed “Bemo,” a desktop assistant robot leveraging LangChain and Tavily integrations for multimodal language understanding.
- Used OpenCV and NLP libraries to process user inputs and context for intelligent task management.
- Integrated Google Gemini LLM to enhance conversational abilities and reasoning.
- Built modular FastAPI endpoints to orchestrate robot actions and AI backend services.

Face Analysis & C++ Recognition System — Python, C++, LFW Dataset

Jun 2025 – Jun 2025

- Benchmarked 11 face-analysis models (6 detectors, 2 landmarks, 3 recognizers) on LFW—achieved 99–100% detection accuracy and 98.6% recognition with a 0.16s/image pipeline.
- Built an optimized C++ face recognition system using MediaPipe and TensorFlow Lite, designed for efficient edge inference on Raspberry Pi.

Lip Reading System — Python, 3D-CNN, OpenCV

Sep 2023 – Oct 2023

- Created a lip-reading system using 3D convolutional neural networks to transcribe silent video lip movements into text.
- Trained on a curated dataset and achieved strong word-level classification accuracy.

Mimic-NAO: Human-to-Robot Pose Control — NAOqi, Cartesian Inverse Kinematics

Jan 2023 – Feb 2023

- Built a real-time robot mimicry system capturing 3D human pose via MediaPipe and mapping it to NAO robot using Cartesian inverse kinematics through NAOqi.
- Implemented closed-loop control reproducing head pitch/yaw and limb movement with 96%+ accuracy, using analytical and Jacobian-based kinematics chains.

Cubot: Rubik's Cube Solver Robot — Python, OpenCV, Arduino

Sep 2022 – Jan 2023

- Designed a Rubik's Cube solving robot: OpenCV-based color detection, Kociemba solving algorithm, and Arduino motor actuation.

CERTIFICATES & ACHIEVEMENTS

Qualified to ACPC Regionals (2024) and received the **First Solve Award** at ECPC 2024 Qualifiers, ranking among the **Top 50 Teams** in the Arab region.

Placed **Top 10** during ECPC Qualification Day (2023, 2024); Achieved **1st Place** in the ECPC College Qualifiers in both years.

RoboCup@Home EDU Workshop, Feb–Mar 2023: Focused on vision, robotics, and autonomous interaction.

DEBI Robotics Challenge, Ministry of Communications, Jul–Aug 2022: Used TurtleBot3 for motion planning, ball detection, and obstacle avoidance.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, Dart, JavaScript, HTML/CSS

Platforms: Arduino, AWS, Linux, Windows

Tools/Libraries: TensorFlow, Pandas, NumPy, OpenCV, Scikit-learn, Node.js, ROS, MATLAB, Flask, Git, Docker, Flutter, MySQL, Eclipse, VS Code, Clion

LEADERSHIP & VOLUNTEERING

Judge, RoboCup Egypt Nationals – Rescue Maze League (2025): Evaluated teams based on mission success, strategy, and robot performance.

Founder & Coach, Competitive Programming (CP) Communities – AAST Alamein & Assiut University: Created CP groups, delivered training in algorithms, DS, and contest preparation.

Team Member, Sky Experts Drones Software Team – Jan 2024 - Sep 2024: Developed control algorithms for a quadcopter using ROS and contributed to mission planning and strategy in the computer vision team.